

Sweep Velocity Tradeoff Model

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Introduction

Every pitcher throws a breaking ball that is slower than their fastball. That's by design. But the size of that velocity gap varies enormously across pitchers and pitch types, and not all that variation can be explained by the physical characteristics of the pitch itself. Some pitchers throw breaking balls that are faster than their shape would predict, while others leave velocity on the table relative to their movement profile. The question I explore here is whether this velocity gap, “delta”, matters for in game pitch effectiveness.

To answer this question, I built a model in R that predicts the expected velocity drop between a pitcher's fastball and each of their breaking balls using movement profile, spin characteristics, and release data. The residual, the part of the velocity gap that shape cannot explain, is then tested against run value per 100 pitches (RV/100) from Baseball Savant. My hypothesis was that pitchers whose breaking balls are faster than their shape would predict would generate better outcomes on those pitches, showing up in a higher RV/100 value.

Data

The model uses pitch-level Statcast data from 2020-2024, totaling roughly 3.1 million pitches. I aggregated all pitches to one row per pitcher, per pitch type, per season. Changeups, screwballs, knuckleballs, eephus pitches, and unknown pitch types are excluded because they are not “breaking balls.”

For this project, my target variable is delta: the velocity difference between a pitcher's breaking ball and their weighted average fastball velocity (combining four-seamers and sinkers by pitch count). A less negative delta means the breaking ball is faster relative to the fastball than expected.

To draw correlation to an in-game metric, I used sourced RV/100 by pitcher and pitch type for 2023-2024. Validation is restricted to 2023-2024 because that is when the sweeper (SW) was introduced as a distinct pitch classification in 2023. Pre-2023, sweepers were coded as sliders, which would contaminate the pitch-level join for all slider subtypes. The model itself still trains on the full 2020-2024 window.

Feature Engineering

One important learning moment for me, and a crucial model design choice involved spin efficiency. I initially derived a velocity-based spin efficiency measure using the formula $(\text{active_spin} / \text{raw_spin}) * 100$, where active_spin itself depends on pitch velocity. This created a leakage problem because delta, the target variable, is a function of velocity which gives each model a shortcut. I caught this during diagnostics when the model performed suspiciously well. The fix was to replace it with a trig-based proxy derived from spin direction, which preserves the

meaningful information about how efficiently the ball spins without leaking velocity data into the feature set.

Slider Clustering

Sliders are not a homogeneous pitch category. A sweeper with 17 inches of horizontal break behaves fundamentally differently from a tight gyro slider. Rather than treat all sliders as one group, I used k-means clustering ($k=3$) on horizontal break, IVB, and spin rate to separate sliders into three movement-based subtypes: Sweeper, Gyro, and Traditional Slider.

The k-means label was stored in a separate column (`pitch_subtype`) and never overwrote the original Statcast `pitch_type` code preserving the correct join key for the external validation data. If I had overwritten `pitch_type` with the cluster label, the entire validation pipeline would have broken because there would be no clean key to match against the Savant arsenal CSVs.

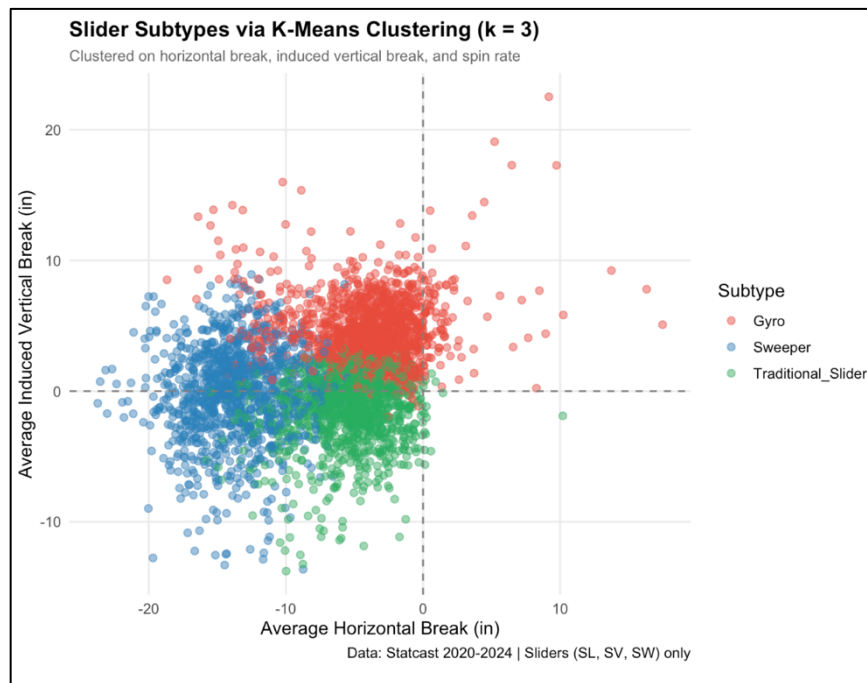


Figure 1: Slider Distribution by Subtype

Model Structure

Four models of increasing flexibility were trained and compared on the same cross-validation splits: linear regression (LM), a generalized additive model (GAM), random forest (RF), and XGBoost. All four used the same 8 predictor variables:

- Average Spin (RPM)
- Inducted Vertical Break
- Horizontal Break

- Release Extension
- Release Height
- Average Fastball Velocity (MPH)
- Spin Direction (Measured in Radians)
- Spin Efficiency (Measured in Radians)

These features describe the shape of each pitch. These features teach the model what each pitch physically looks like, without disclosing breaking ball velocity. Each model learns what velocity gaps each shape profile should produce, which means the residual captures whatever the shape profile can't explain.

Cross-Validated RMSE by Model		
5-fold grouped CV by <i>pitcher_id</i>		
Model	CV RMSE	Eval
RF	1.91	CV
GAM	1.92	CV
XGB	1.92	CV
LM	1.97	CV

Figure 2: Cross Validated RMSE Comparison

Cross Validation Design

The cross-validation uses 5-fold grouped splits, where folds are grouped by *pitcher_id*. This means no pitcher appears in both the training and test set within the same fold. Without grouping, the model could memorize pitcher-specific patterns during training and artificially deflate the CV error by predicting on pitchers it already learned. Grouping by pitcher forces the model to generalize to entirely unseen pitchers in each fold.

Out-Of-Fold Predictions

Rather than using a single train/test split, which would produce residuals for only ~10% of pitchers, the cross-validation loop saves held-out predictions for every pitcher across all 5 folds. Upon completion, every pitcher in the dataset has been held out exactly once, producing a predicted delta for 100% of the population.

The residual is calculated as: $\text{Residual} = \text{Actual delta} - \text{expected delta}$

A positive residual means the breaking ball is faster than the model predicts given its shape. A negative residual means it is slower than expectation.

Arsenal Example

To make the model tangible, the table below shows Dylan Cease's 2024 arsenal with actual vs expected velocity drop. Cease is a useful example because his breaking balls all land within roughly 1 mph of the model's expectation, meaning his velocity gaps are well explained by his pitch shape.

Dylan Cease 2024 Pitch Arsenal Breakdown							
Actual vs Expected Velo Drop							
Pitch	Count	Velo	Velo Drop	Exp. Velo Drop	HB	VB	Spin Eff.
FF	1388	96.9	NA	NA	2.5	18.4	91.3
SL	1371	87.7	-9.2	-8.4	-2.2	1.9	11.8
CB	251	82.3	-14.6	-15.1	-3.1	-15.8	63.3
SW	131	82.6	-14.3	-13.9	-11.2	-11.1	58.7
CT	19	91.8	-5.1	-5.6	-3.0	6.1	30.6

Figure 3: Cease Arsenal Review

Validation Against RV/100

In order to drive decision making, it is important to consider whether this residual is correlated to in game results. To test this, I joined the calculated out-of-fold residuals to Baseball Savant's RV/100 data on *pitcher_id*, *season*, and *pitch_type*. Each row in the correlation analysis represents one pitcher's pitch type in one season. I computed Pearson r between residual and RV/100 within each pitch type and across the full dataset. Scatter plots are faceted by Savant pitch type (Sweeper, Slider, Curveball, Cutter) so every dot in a panel joined to the same external data.



Figure 4: Residual vs. RV Per Pitch

I began testing with a minimum pitch count threshold of 30 pitches. At this threshold, the overall Pearson r was 0.014 ($p = 0.50$), which I initially interpreted as a null result. Raising the threshold to 100 pitches removed noisy small-sample entries and revealed a weak but statistically significant correlation of 0.065 ($p = 0.01$) across 1,547 pitcher-pitch-season observations. Sweepers showed the strongest signal at $r = 0.199$, while cutters, sliders, and curveballs ranged from 0.037 to 0.053. The relationship is weak and only becomes visible once small samples are filtered out.

Residual vs. RV/100 Correlation by Pitch Type		
<i>r > 0: faster-than-expected breaking balls produce better run value</i>		
Pitch Type	N	Pearson r (Residual x RV/100)
Sweeper (SW)	261	0.199
Cutter (CT)	311	0.053
Slider (SL)	627	0.038
Curveball (CB)	348	0.037
Overall (All Pitches)	1,547	0.065
<i>OOF XGB residuals, 5-fold grouped CV Savant RV/100 2023-2024 Min 100 pitches</i>		

Figure 5: Correlation Summary

Interpreting Results

There are a few possible explanations for why the residual only carries minimal predictive signal:

1. The initial framing of the question assumes that breaking balls with positive residual should produce better outcomes due to deception and “stuff”. But there is also a credible argument in the opposite direction, where a larger velocity gap between fastball and breaking ball creates more timing disruption, meaning a slower-than-expected breaking ball could be more effective. If both effects exist and partially cancel, the net correlation would be minimal, which is what the data shows.
2. Run value per 100 pitches is an aggregate outcome metric that includes contact quality, sequencing effects, count leverage, and batter quality. If the residual captures something real about pitch deception, it might show up in more direct swing-decision metrics (chase rate, whiff rate) without surfacing in total run value.

Leaderboards

Outperformers are pitchers whose breaking balls are consistently faster than expected. Laggards are pitchers whose breaking balls are consistently slower. The laggards are particularly interesting from a player development perspective, as this gap might represent an opportunity for additional velocity. While there is no general correlation between residual and RV/100, a pitcher-specific approach can be more appropriate where a front office can identify a laggar with a poor RV/100 on a specific pitch could benefit from exploring whether more velocity is physically available.

Breaking Ball Velocity Outperformers						
Positive residual = actual velo drop smaller than expected						
#	Pitcher	Season	Pitch	Pitches	Residual (mph)	RV/100
1	Shane Bieber	2023	CB	269	6.22	−0.80
2	Jose Soriano	2023	CB	278	5.98	1.50
3	Brendon Little	2024	CB	221	5.49	1.00
4	Craig Kimbrel	2023	CB	366	5.34	0.70
5	Emmanuel Clase	2023	CT	758	5.21	0.70
6	Gus Varland	2023	Traditional_Slider	181	5.03	−2.50
7	Marcus Stroman	2024	Traditional_Slider	449	5.02	−2.30
8	Matt Barnes	2024	CB	103	4.87	−1.00
9	Luke Jackson	2023	Traditional_Slider	367	4.78	1.40
10	Tommy Nance	2024	CB	121	4.70	1.30
11	Emmanuel Clase	2024	CT	774	4.68	3.10
12	Kyle Bradish	2023	Gyro	824	4.68	1.50
13	Cory Abbott	2023	CB	332	4.65	−0.10
14	Luke Jackson	2024	Traditional_Slider	489	4.64	−0.60
15	Matt Moore	2024	CB	150	4.58	−0.20

Figure 6: Residual Leaders

Breaking Ball Velocity Ladders						
Negative residual = actual velo drop larger than expected						
#	Pitcher	Season	Pitch	Pitches	Residual (mph)	RV/100
1	Eury Perez	2023	CB	215	-7.07	1.70
2	Rich Hill	2023	Gyro	174	-6.40	-1.80
3	Nathan Eovaldi	2024	CB	358	-5.30	-1.10
4	Freddy Peralta	2023	CB	355	-5.26	0.50
5	David Bednar	2024	CB	209	-5.14	0.10
6	Bowden Francis	2023	CB	183	-4.97	2.30
7	Yu Darvish	2024	CB	172	-4.95	1.00
8	Chris Bassitt	2024	Gyro	217	-4.87	1.00
9	Chris Flexen	2024	CB	342	-4.86	-0.10
10	Chris Bassitt	2023	Gyro	282	-4.80	0.10
11	Nathan Eovaldi	2023	CB	301	-4.72	0.90
12	Jose Urena	2024	Sweeper	586	-4.64	0.10
13	Johan Oviedo	2023	CB	445	-4.63	0.90
14	Freddy Peralta	2024	CB	213	-4.50	0.80
15	Cristian Javier	2023	CB	217	-4.42	-1.00

Figure 7: Residual Ladders

Curveballs dominate both leaderboards because they have the largest velocity gap from the fastball, which creates more room for the actual delta to deviate from expectation in either direction. It is also worth noting that a pitcher on the ladders table is not necessarily underperforming. Their breaking ball is simply slower than expected given its shape, which could be an intentional pitch design choice.

Limitations and Next Steps

Given this was my first attempt at a baseball research project, there are several limitations I would want to address in the future.

1. Automated feature selection, like a Boruta, would provide rigorous justification for variable inclusion and could reveal interactions not identified by manual testing.
2. XGBoost hyperparameters were tuned through continuous testing rather than through grid search. Systematic optimization could improve efficiency on the margins.

The next step I would take is validating the residual against swing decision metrics like chase and whiff rates rather than Run Value.